Lab 7: Fuzzing for Input Validation Bugs

Bookstore Trade-In (processor.py) - Hypothesis fuzz testing

Fixes made: all three functions now validate their input and return None instead of raising for non-strings, empty or malformed text, a missing or extra '/', nan/inf, a zero or negative lifespan, and unknown condition codes.

Step 2: Run the buggy application

python processor.py with the original script. An empty ISBN input crashes with IndexError: string index out of range.

```
PS C:\Users\rchuk\L6_L7_09_bookstore_trade_in\Lab 7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2):
Traceback (most recent call last):
  File "C:\Users\rchuk\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 40, in <module>
    main()
    ~~~~^^
  File "C:\Users\rchuk\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 29, in main
    print(verify_isbn10_format(isbn_in))
          ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^
  File "C:\Users\rchuk\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 3, in verify_isbn10_format
    first_digit = int(cleaned[0])
                      ~~~~~~~^^^
IndexError: string index out of range
```

Step 4: Run test_processor.py against the buggy script

Hypothesis finds failing inputs such as None, empty string, '/', '0/0' and unknown condition codes. Result: 8 failed, 4 passed.

```
=========================================== short test summary info ===========================================
FAILED test_processor.py::test_verify_isbn10_format_no_crash - ExceptionGroup: Hypothesis found 4 distinct failures. (4
sub-exceptions)
FAILED test_processor.py::test_calculate_depreciation_factor_safe - ExceptionGroup: Hypothesis found 3 distinct failures
. (3 sub-exceptions)
FAILED test_processor.py::test_lookup_condition_multiplier_safe - ExceptionGroup: Hypothesis found 2 distinct failures.
(2 sub-exceptions)
FAILED test_processor.py::test_all_functions_handle_non_string_input - AssertionError: verify_isbn10_format crashed with
: AttributeError("'NoneType' object has no attribute 'replace'") f...
FAILED test_processor.py::test_depreciation_result_is_none_or_within_bounds - ExceptionGroup: Hypothesis found 3 distinc
t failures. (3 sub-exceptions)
FAILED test_processor.py::test_condition_result_is_none_or_known_multiplier - ExceptionGroup: Hypothesis found 2 distinc
t failures. (2 sub-exceptions)
FAILED test_processor.py::test_isbn_result_is_none_or_formatted_string - ExceptionGroup: Hypothesis found 4 distinct fai
lures. (4 sub-exceptions)
FAILED test_processor.py::test_edge_cases_do_not_crash - IndexError: string index out of range
========================================= 8 failed, 4 passed in 2.74s =========================================
```

Step 5: Run the fixed application

python processor.py with the fixed script. Empty ISBN, 5/0 and BROKEN are all handled with clear messages and no crash.

```
PS C:\Users\rchuk\L6_L7_09_bookstore_trade_in\Lab 7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2):
Invalid ISBN-10

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): 5/0
Invalid age/lifespan input

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR): BROKEN
Unknown condition code
```

Step 6: Re-run test_processor.py against the fixed script

All 12 tests pass.

```
PS C:\Users\rchuk\L6_L7_09_bookstore_trade_in\Lab 7> python -m pytest test_processor.py
============================================== test session starts ==============================================
platform win32 -- Python 3.14.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\rchuk\L6_L7_09_bookstore_trade_in\Lab 7
plugins: hypothesis-6.170.15, platformdirs-4.13.7
collected 12 items

test_processor.py ............                                                                    [100%]

============================================== 12 passed in 0.74s ===============================================
```